

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459131.80018 +/- 0.00017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.012
Transit depth (Rp/Rs)^2	3.6e-06 +/- 0.00045 [%]
Orbital Inclination (inc)	85.22 +/- 1.42
Airmass coefficient 1 (a1)	1.2391 +/- 0.0096
Airmass coefficient 2 (a2)	-0.1513 +/- 0.0033
Scatter in the residuals of the lightcurve fit is	0.76 %
Best Comparison Star	None
Optimal Aperture	5.13
Optimal Annulus	11.55
Transit Duration (day)	0.0711 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.012 vs 0.1301 (deviation 8.64σ)
Duration fit vs published	0.0711 d vs 0.0908 d (deviation 2.62σ)
Mid-time O–C	0.3 min
Depth SNR	0.0
Residual scatter	0.76 %

Light curve

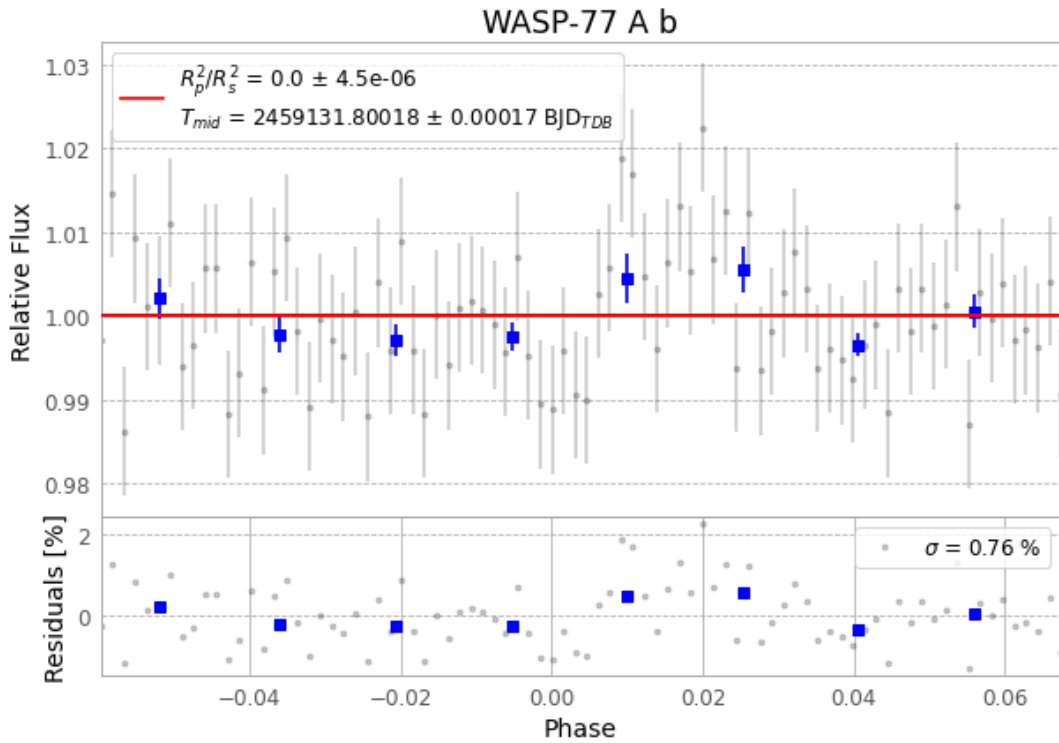


Figure 1 — FinalLightCurve_WASP-77 A b_2020-10-08.png (SHA-256 8f1ca7e7e6634a57...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [466, 150], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4227f435df506148...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

84 FITS frames (55 MB), WASP-77201009050615.FITS → WASP-77201009091512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-201009.log (52f1cab54ba...), FinalLightCurve_WASP-77 A b_2020-10-08.png (8f1ca7e7e663...), FinalLightCurve_WASP-77 A b_2020-10-08.csv (4c606683dfc8...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 07:29Z.